■ Official University Master Textbook

Chapter 1: Electrical Power Engineering & Smart Grid Fundamentals from First Principles

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Section 1: Fundamental Electrical Physics (Voltage, Current, Power)

1.1 Definition of Electricity & Charge

Electricity is the set of physical phenomena associated with the presence and motion of matter that has a property of electric charge. The fundamental unit of charge is the electron, carrying -1.602 x 10^-19 Coulombs. Electric current (I) is the time rate of flow of electric charge: I = dQ / dt, measured in Amperes (Coulombs per second). Voltage (V) is the electric potential difference or electromotive force (EMF) that pushes electrons through a conductor, measured in Volts (Joules per Coulomb).

1.2 Ohm's Law and Electrical Resistance

Ohm's Law states that the current flowing through a conductor between two points is directly proportional to the voltage across the two points: V = I * R, where R is resistance in Ohms. Resistance is determined by resistivity (rho), length (L), and cross-sectional area (A): R = rho * (L / A). In AC systems, opposition to current flow is called Impedance (Z), combining Resistance (R) and Inductive/Capacitive Reactance (X): Z = R + jX.

Section 2: Alternating Current (AC) Power Fundamentals

2.1 AC Sine Waves & RMS Voltage

Unlike Direct Current (DC) which flows in one direction, Alternating Current (AC) continuously reverses direction in a sinusoidal waveform: v(t) = V_peak * sin(2 * pi * f * t). In India and Europe, grid frequency (f) is 50.0 Hz (reversing 50 times per second). Root-Mean-Square (RMS) voltage represents the equivalent DC heating voltage: V_rms = V_peak / sqrt(2) = 230V for single phase, 400V for 3-phase line-to-line.

2.2 Power Triangle (Active, Reactive, Apparent Power & Power Factor)

In AC power systems with inductive loads (motors, transformers), current lags behind voltage by phase angle (phi). Power is split into three components:

- **Active Power (P)**: Real work consumed, measured in Kilowatts (kW). P = V * I * cos(phi).
- **Reactive Power (Q)**: Magnetic field energy oscillating back and forth without doing work, measured in kVAR. Q = P * tan(arccos(phi)).
- **Apparent Power (S)**: Total vector power capacity, measured in kVA. S = sqrt(P^2 + Q^2) = V * I.
- **Power Factor (cos phi)**: Energy efficiency ratio (P / S). Power companies require cos phi >= 0.90 to prevent grid line losses.

Section 3: Distribution Network Architecture & Transformers

3.1 Grid Architecture (Generation -> Transmission -> Distribution)

1. **Generation**: Hydro, thermal, or solar plants generate power at 11kV to 25kV.
2. **Transmission**: Step-up transformers raise voltage to 110kV - 765kV to minimize I^2 * R heat losses over long distances.
3. **Substation**: Step-down transformers reduce voltage to 11kV distribution feeders.
4. **Distribution Feeder**: 11kV radial lines carry power down streets to local distribution transformers.
5. **Secondary Distribution**: Step-down transformers reduce 11kV to 400V 3-phase / 230V single-phase to power homes.

Section 4: Grid Short-Circuit Faults & Relay Protection

4.1 Types of Short-Circuit Faults

• **3PH (Three-Phase Symmetrical)**: Severe fault where all 3 lines short-circuit together. I_f = V_ph / |Z1 + Rf|.
• **SLG (Single Line-to-Ground)**: Most common fault (70-80% of events) where 1 phase touches ground. I_f = 3 * V_ph / |2*Z1 + Z0 + 3*Rf|.
• **LL (Line-to-Line)**: Unsymmetrical fault between 2 phases. I_f = sqrt(3) * V_ph / |2*Z1 + 2*Rf|.

4.2 Overcurrent Protection Relay Trip Curve (IEC Extremely Inverse)

Protection relays isolate short circuits before equipment explodes. Under IEC 60255 standards, trip time (t) follows the Extremely Inverse curve: t = 80 / ((I_fault / I_pickup)^2 - 1) seconds, ensuring higher fault currents trip the relay faster (in milliseconds).